

Use-Case Flow Specification(Problem Statement – 44)

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Use Case ID / Name	UC-03: Identify Missing Index Candidates
Primary Actor(s)	Database Administrator (initiates review); System (automated analysis)
Secondary Actor(s)	Backend Lead (notified of findings via digest, UC-05)
Trigger	A new slow query log batch has been ingested and parsed (includes UC-02: Parse Execution Plan), or the Database Administrator manually requests an on-demand analysis.
Preconditions	1. The target query's execution plan has already been successfully parsed (UC-02). 2. The database schema metadata (existing indexes, table sizes, column statistics) for the affected table is available and less than 24 hours stale. 3. The Database Administrator has an active, authenticated session with read access to the relevant database connection.
Postconditions	Success: One or more ranked index-candidate recommendations (with target table, column(s), index type, and estimated impact) are persisted and linked to the source query; the recommendation status is set to "Pending". Failure: No recommendation is created, and the query is flagged as "Analyzed - No Action" so it is not re-processed unnecessarily.

Main Success Scenario

- 1.The System retrieves the parsed execution plan for the query (produced by UC-02) together with current table and index metadata for the referenced tables.
2. The System scans the plan for sequential (full table) scans, filters, and joins performed on tables above a configurable row-count threshold.
3. The System cross-references the joined columns against existing indexes to determine which columns lack adequate index coverage.
4. The System ranks candidate columns by estimated performance impact (using scan cost and query frequency) and drafts a composite index definition for the highest-impact candidate.
5. The System persists the recommendation with status "Pending" and links it to the source query and execution plan.
6. The System notifies the Backend Lead and Database Administrator that a new recommendation is available.
7. The Database Administrator reviews the recommendation, and the use case ends successfully.

Alternate Flow

A1. No Viable Index Candidate Found (branches after step 3)

- A1.1. The System determines that all filtered/joined columns already have adequate index coverage, or that the query pattern is too irregular to justify a new index (e.g., highly selective ad-hoc filters).
- A1.2. The System marks the query as "Analyzed - No Action" instead of creating a recommendation.
- A1.3. The System logs the analysis outcome and reasoning for audit purposes.
- A1.4. The use case ends without producing a recommendation (postcondition: failure branch).